

Extra Exercise on Finding Area using Parametric Equations

Find the area of the regions bounded by the parametric curves.

(1) $x = 2 \sin^2 \theta$, $y = 2 \sin^2 \theta \tan \theta$, $0 \leq \theta \leq \frac{\pi}{2}$.

(2) $x = 2 \cot \theta$, $y = 2 \sin^2 \theta$, $0 \leq \theta \leq \pi$.

(3) $x = 2a \cos t - a \cos(2t)$, $y = 2a \sin t - a \sin(2t)$, $0 \leq t < 2\pi$.

(4) $x = a \sin(2t)$, $y = b \sin(t)$, $0 \leq t < 2\pi$.

(5) $x = 2a \cos t - a \sin(2t)$, $y = b \sin t$, $0 \leq t < 2\pi$.

(6) Region bounded by $x = 3 - t^2$, $y = t^2 + 3t$ and y -axis.

The set up for the expression of integral is given on the next page.

- (1) Type of integral: $\int y \, dx$
 Starting point: $(0, 0)$
 Corresponding θ : $\theta_0 = 0$
 Ending point: $(2, \infty)$
 Corresponding θ : $\theta_1 = \pi/2$
 The expression of Area: $\int_0^{\pi/2} (2 \sin^2 \theta \tan \theta) (4 \sin \theta \cos \theta) \, d\theta$
- (2) Type of integral: $\int y \, dx$
 Starting point: $(-\infty, 0)$
 Corresponding θ : $\theta_0 = \pi$
 Ending point: $(\infty, 0)$
 Corresponding θ : $\theta_1 = 0$
 The expression of Area: $\int_{\pi}^0 (2 \sin^2 \theta) (-2 \csc^2 \theta) \, d\theta$
- (3) Type of integral: $\int x \, dy$
 Starting point: $(a, 0)$
 Corresponding t : $t_0 = 0$
 Ending point: $(a, 0)$
 Corresponding t : $t_1 = 2\pi$
 The expression of Area: $\int_0^{2\pi} (2a \cos t - a \cos(2t)) (2a \cos t - 2a \cos(2t)) \, dt$
- (4) Type of integral: $\int x \, dy$ (Split into two regions with same region, we find the region of upper loop)
 Starting point: $(0, 0)$
 Corresponding t : $t_0 = 0$
 Ending point: $(0, 0)$ (after one loop)
 Corresponding t : $t_1 = \pi$
 The expression of Area: $2 \int_0^{\pi} (a \sin(2t)) (b \cos t) \, dt$
- (5) Type of integral: $\int x \, dy$
 Starting point: $(2a, 0)$
 Corresponding t : $t_0 = 0$
 Ending point: $(2a, 0)$
 Corresponding t : $t_1 = 2\pi$
 The expression of Area: $\int_0^{2\pi} (2a \cos t - a \sin(2t)) (b \cos t) \, dt$
- (6) Type of integral: $\int x \, dy$
 Starting point: $(0, 3 - 3\sqrt{3})$
 Corresponding t : $t_0 = -\sqrt{3}$
 Ending point: $(0, 3 + 3\sqrt{3})$
 Corresponding t : $t_1 = \sqrt{3}$
 The expression of Area: $\int_{-\sqrt{3}}^{\sqrt{3}} (3 - t^2) (2t + 3) \, dt$